AIDCF-018-RES Resilience, Continuity and Recovery Domain Guide

AIDCF – AI Data Centre Facilities Assurance Framework

Document control

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Revision history

Version	Date	Summary
1.0	2026-10-10	First issue of the RES domain guide, aligned to the rewritten MCF-RES control records, per-control gate mapping and canonical vocabularies.

1 Purpose and scope

This guide explains how the Resilience, Continuity and Recovery (RES) domain is assessed. RES is the integrating domain of AIDCF: it does not own power, cooling, compute, storage or network systems, but it owns the questions that cut across them. What minimum outcome must survive? Where are the fault domains and how far can a failure spread? What happens when capacity is partially lost? How is the capability restored, in what order and how fast? Has any of this been demonstrated?

The guide is informative except where a statement carries a DG-RES-REQ identifier. The 24 RES controls in AIDCF-007A are normative; if this guide and a control record disagree, the control record prevails and the discrepancy shall be raised as a baseline issue.

In scope:

- Minimum service outcomes, impact tolerances and business-impact analysis for each Capability Instance.
- Graceful degradation: planned reduction of AI service under partial loss.

- Scenario libraries, site fault-domain models, containment boundaries, blast-radius analysis and isolation validation.
- Continuity strategy, alternate capacity, site and provider continuity, and crisis priorities.
- Recovery dependencies (including circular and cold-start dependencies), black-start and restoration sequencing across power, cooling, network, storage, compute and control plane, configuration restoration, and data and checkpoint recovery.
- Component tests, integrated scenario tests, tabletop exercises and the lessons-and-retest loop.
- Crisis command, stakeholder communications, emergency change and post-event adaptation.

Domain-specific recovery mechanics remain in their domains: POW-024 (electrical black-start and restoration), THM-024 (thermal recovery), CMP-023 (node rebuild), STO-019 (storage recovery sequencing), NET-015 (OOB resilience). RES assesses whether these compose into an end-to-end, tested restoration of the Service Outcome.

2 Why this domain matters for AI capability

AI infrastructure changes the resilience problem in four ways.

1. **Concentration.** A single training job can occupy tens of thousands of accelerators in one hall or campus. A failure that a conventional data centre would absorb as the loss of a few tenants' racks can halt the whole Capability Instance. Blast radius must therefore be expressed in capability terms (accelerator-hours, jobs, tokens per second), not in racks or kilowatts.
2. **Tight coupling.** Synchronous training means the failure of any participant stops progress for all participants. Resilience for training is largely achieved not by keeping every component running but by **checkpointing** and **fast restart**: the impact of a failure is the work lost since the last checkpoint plus the time to detect, replace or exclude the failed component and restart. Lost work is a first-class impact measure.
3. **Power and thermal dynamics.** AI loads can change by large fractions of rated capacity in seconds as jobs start, synchronise and checkpoint. Restoration after a site event must manage load steps against generator and UPS capability and must not bring heat load online before heat removal is re-established. Liquid-cooled systems have limited thermal ride-through, so cooling restoration and IT load sequencing are coupled.
4. **Digital control-plane dependency.** Restoration depends on identity, time, DNS, orchestration, image repositories and licence services that are frequently hosted on the very infrastructure being restored. These circular dependencies are among the most common reasons recovery plans fail on first real use.

RES controls therefore provide most of the CRITICAL evidence for **CG-09 Failure containment** and **CG-10 Recovery readiness**, and they supply the tolerances against which every other domain's resilience claims are judged.

3 Reference patterns and variants

3.1 Resilience strategies

Strategy	Typical use	Key RES evidence
In-site redundancy with checkpoint restart	Large single-site training	Fault-domain model, checkpoint policy, restart tests
Active-active multi-site	Latency-tolerant inference serving	Alternate capacity tests, DCI diversity, traffic failover
Warm standby capacity	Priority fine-tuning or sovereign workloads	Relocation tests, image and data compatibility

Strategy	Typical use	Key RES evidence
Contracted burst capacity (cloud)	Overflow and continuity for inference or smaller training	Contract terms, activation tests, data transfer analysis
Recovery-capacity pool (AIDCF-013)	Shared reserve across Capability Instances	Priority scheme, reservation enforcement
Accepted suspension	Non-critical research training	Approved acceptance, tolerance justification

3.2 Impact tolerance framing

Impact tolerance is the maximum disruption a Capability Instance can sustain before the consequence becomes intolerable to its stakeholders. Financial-sector operational resilience regimes (for example the UK important-business-services and impact-tolerance approach, and the EU Digital Operational Resilience Act) use similar framing; AIDCF applies it to AI capability. Typical dimensions:

- **Maximum tolerable disruption** (duration).
- **Minimum retained capacity** in DEGRADED or EMERGENCY state.
- **Maximum lost work** (training) or **maximum failed requests** (inference).
- **Recovery time objective (RTO)** and **recovery point objective (RPO)**; for training, RPO is effectively the checkpoint interval plus any un-replicated checkpoints.
- **Integrity tolerance**: zero tolerance for silent model or data corruption is typical; recovery must restore a verified state.

3.3 Restoration architecture

The canonical recovery chain in AIDCF-013 is *detection → decision → containment → alternate capacity → restore → validate*. For a site-wide event, the restoration layer order is:

1. **Power**: utility return or generator start; distribution re-energisation in controlled steps; UPS recharge management.
2. **Cooling**: heat rejection, facility water and chilled water, then CDUs and secondary loops, confirmed stable before IT load.
3. **Out-of-band and core network**: OOB first (it is the tool for everything else), then core and management networks.
4. **Core services**: time, DNS, DHCP, IPAM, AAA and identity, ideally from fallbacks that do not depend on the main cluster.
5. **Storage**: metadata services and quorum before data services; integrity checks before admitting clients.
6. **Compute**: BMC reachability, firmware validation, health checks and burn-in before admission; staged power-on to respect load-step limits.
7. **AI control plane**: cluster manager, scheduler, image registry, then workload admission by priority.
8. **Validation**: Service Outcome checks (benchmarks, test jobs, inference canaries), not only system availability.

4 Required functions and capability linkage

Required Function	Principal RES controls
Define minimum outcome and impact tolerance	RES-001, RES-012
Degrade gracefully under partial loss	RES-002
Anticipate credible scenarios	RES-003
Own resilience across boundaries	RES-004

Required Function	Principal RES controls
Bound failure propagation	RES-005, RES-006, RES-007, RES-008
Sustain or resume the outcome elsewhere	RES-009, RES-010, RES-011
Restore the outcome within tolerance	RES-013, RES-014, RES-015, RES-016
Demonstrate and improve resilience	RES-017, RES-018, RES-019, RES-020
Command and adapt during and after crises	RES-021, RES-022, RES-023, RES-024

DG-RES-REQ-001 An assessor shall not evaluate any RES-02 to RES-05 control as CONFORMING for a Capability Instance whose minimum service outcome and impact tolerance (RES-001) are absent or unapproved; the affected results shall be recorded as UNKNOWN with a US5 UNKNOWN Record against RES-001.

5 Dependencies, fault domains and common-mode concerns

5.1 The site fault-domain model

RES-005 requires a single fault-domain model covering electrical distribution paths and UPS systems, cooling strings and CDU groups, fire compartments, water zones, network planes, compute pods, storage failure domains and control-plane clusters, reconciled to as-built evidence. Its value is in the **overlaps**: redundant network spines inside one UPS output, storage replicas in one fire compartment, control-plane quorum members in one rack, both CDUs of a pair on one electrical board. The model is most efficiently held as a graph using AIDCF-014 classes (FaultDomain, Dependency, InfrastructureSystem, AssetInstance) and SHARES_FAULT_DOMAIN_WITH relationships, so that redundancy claims can be tested by query.

5.2 Recovery dependencies and circularity

Recovery dependencies are often different from operating dependencies. Examples that RES-013 should surface:

- Identity provider hosted on the cluster it unlocks (AIDCF-004 X19).
- DNS required to resolve the time server, while certificates fail until time is correct.
- Node images and firmware bundles stored only on the failed storage system.
- Licence servers for storage or orchestration software on the failed cluster.
- BMS/EPMS historian or engineering workstation virtualised on the failed IT platform.
- Runbooks and contact lists held only in a corporate wiki behind the failed identity provider.
- Key individuals for whom no alternate exists (shared human dependency, DEP-008).

Each cycle needs a documented break: an offline copy, a local fallback, an out-of-band path or a manual procedure, stored outside the failure domain it serves.

5.3 Common-mode concerns specific to resilience

- **Common-mode in recovery**: the same automation, image or configuration baseline used to restore every node can reintroduce a defect everywhere (X15). Recovery must restore a verified, pre-incident baseline.
- **Common-mode in alternate capacity**: a cloud burst contract offers little continuity if the regional cloud and the primary site share grid, carrier or weather exposure, or if the alternate is oversubscribed during the same event.
- **Common-mode in people**: crisis command, facility engineering and platform operations may all depend on a small group of individuals; exercises reveal this.

6 Threats, hazards and failure scenarios with propagation paths to Service Outcomes

RES uses the full AIDCF-004 scenario library. The cross-domain scenarios most often decisive for RES gate results are below.

Scenario	Path summary	Service Outcome effect	RES controls exercised
X01 Utility loss + generator no-start + UPS autonomy exhaustion	Utility loss → UPS carries load → one generator fails to start → autonomy consumed → load shed or drop	Training halts; lost work equals time since last checkpoint; inference availability depends on shedding	RES-002, RES-012, RES-014, RES-016, RES-018
X02 Cooling loss at sustained AI load	Chiller or CDU failure → supply temperature rise → thermal throttling → protective shutdown	Throughput falls then halts; restart constrained by cooling recovery	RES-002, RES-014, RES-018
X04 Maintenance removes hidden shared dependency	Planned isolation of one path → hidden shared element removes the redundant path	Unplanned outage during maintenance	RES-005, RES-007, RES-008
X05 BMS/EPMS compromise or malfunction	Malicious or faulty command → setpoint or mode change → plant response	Thermal or electrical continuity defeated; recovery needs trusted configuration	RES-015, RES-019, RES-021
X08 Firmware rollout common-mode failure	Update pushed broadly → identical failure across domains	Large blast radius; recovery depends on rollback and image availability	RES-007, RES-013, RES-015
X09 Fire disables paired infrastructure routes	Fire in one compartment → paired cable or pipe routes both affected	Loss of both "redundant" paths	RES-005, RES-006, RES-008
X15 Recovery reintroduces corrupt configuration	Restore from drifted or compromised backup	Fault recurs after recovery	RES-015, RES-020
X18 Loss of remote OEM support	Supplier unavailable during incident	Diagnosis and repair delayed	RES-011, RES-013
X19 Shared authentication service failure	Identity provider outage → no administrative access	Recovery blocked	RES-013, RES-014
X20 Compound flood/fire/grid/telecom event	Several initiators coincide	Site loss	RES-009, RES-010, RES-021

Recovery-induced failures deserve specific attention: simultaneous power-on of thousands of nodes can exceed generator load-step capability; restoring IT load before CDUs and secondary loops are primed and stable can trip thermal protection; starting storage data services before metadata quorum can cause inconsistency; and admitting workloads before health checks can propagate faulty nodes into jobs.

7 Safeguards

Safeguard function	RES contribution
Anticipate	Scenario library (RES-003); business-impact analysis (RES-001)
Prevent	Fault-domain design and containment boundaries (RES-005, RES-006)
Detect	Component and integrated tests (RES-017, RES-018) reveal latent failure
Decide	Crisis command and priorities (RES-021, RES-012)
Respond	Graceful degradation (RES-002); emergency change (RES-023)

Safeguard function	RES contribution
Contain	Blast-radius limits and isolation validation (RES-007, RES-008)
Withstand	Continuity strategy and alternate capacity (RES-009, RES-010, RES-011)
Degrade gracefully	Priority preemption, power capping and checkpoint-and-drain (RES-002)
Recover	Recovery dependencies, sequencing, configuration and checkpoint restoration (RES-013 to RES-016)
Adapt	Lessons and retest; post-event adaptation (RES-020, RES-024)

8 Critical-gate linkage

Gate	Name	CRITICAL controls	SUPPORTING controls
CG-01	Capability and outcome definition	—	RES-001, RES-009, RES-012
CG-02	Boundary and architecture integrity	—	RES-005
CG-03	Critical dependency visibility	—	RES-003, RES-005, RES-007, RES-010, RES-011, RES-013
CG-04	Life safety and hazardous energy	—	RES-008
CG-05	Resilient electrical supply	—	RES-014, RES-017, RES-018
CG-06	Thermal continuity	—	RES-014, RES-017, RES-018
CG-07	Critical control-system integrity	—	RES-015
CG-08	Compute, storage and network functional integrity	—	RES-016
CG-09	Failure containment	RES-005, RES-007, RES-008	RES-001, RES-002, RES-003, RES-006, RES-018
CG-10	Recovery readiness	RES-013, RES-014, RES-016, RES-018	RES-001, RES-002, RES-003, RES-009, RES-010, RES-011, RES-012, RES-015, RES-017, RES-019, RES-020, RES-021
CG-13	Responsibility and authority	—	RES-004, RES-019, RES-021, RES-022
CG-14	Change control	—	RES-015, RES-020, RES-023, RES-024

Rationale for the CRITICAL assignments:

- **CG-09**: RES-005 (fault-domain model reconciled to as-built), RES-007 (blast radius quantified against tolerance) and RES-008 (isolation validated) together are the containment argument. If the model is wrong, the blast radius unknown, or isolation untested, containment is asserted rather than demonstrated.
- **CG-10**: RES-013 (recovery dependencies including circularity), RES-014 (tested restoration sequence), RES-016 (checkpoint and data recovery) and RES-018 (integrated scenario tests) are each necessary for recovery readiness. RES-015 (configuration restoration) is SUPPORTING because CG-10 also draws on domain-specific restoration controls (for example NET-022, CMP-016, STO-019) and SEC-024; a gap in RES-015 requires gate-level evaluation of whether those controls compensate.

RES-001 is SUPPORTING for CG-01, CG-09 and CG-10 rather than CRITICAL, because a missing tolerance is handled more strongly: it raises a US5 UNKNOWN (DG-RES-REQ-001), which caps the decision at INCONCLUSIVE regardless of gate results.

9 Capability-profile parameters

All values are illustrative defaults to be set per capability profile with a recorded basis.

Control	Parameter	Unit	Illustrative default	Basis
RES-001	Maximum tolerable disruption (training outcome)	h	Business-impact derived (illustrative 24)	ISO/TS 22317-style impact analysis; capability-specific.
RES-001	Maximum tolerable disruption (inference outcome)	min	Business-impact derived (illustrative 15)	Capability-specific.
RES-001	Maximum lost training progress	h of cluster time	Equal to checkpoint interval plus restart time (illustrative 2)	Links to RES-016.
RES-001	Minimum retained capacity in DEGRADED state	% of nominal	50 (illustrative)	Capability-profile parameter.
RES-002	Time to shed load to residual target	min	Less than UPS autonomy minus margin (illustrative 5)	Derived from POW and THM ride-through.
RES-002	Accelerator power-cap floor	% of TDP	Vendor-supported range (illustrative 60)	Capability-profile parameter; vendor limits apply.
RES-005	Fault-domain model reconciliation age	days	180 (illustrative)	Capability-profile parameter.
RES-007	Maximum blast radius per single credible failure	% of Capability Instance capacity	Impact-tolerance derived (illustrative 25)	Capability-profile parameter.
RES-008	Isolation validation cycle	months	24 (illustrative)	Capability-profile parameter.
RES-010	Alternate capacity activation time	h	Within maximum tolerable disruption (illustrative 8)	Derived from RES-001.
RES-010	Relocation test interval	months	12 (illustrative)	Capability-profile parameter.
RES-011	Provider continuity evidence age	months	12 (illustrative)	Capability-profile parameter.
RES-014	Target black-start-to-service time	h	Within RTO (illustrative 12)	Derived from RES-001.
RES-014	Maximum IT load step during restoration	% of generator rating	Generator- and UPS-specific (illustrative 25)	From POW-020 load-step analysis.
RES-014	Integrated restoration test interval	months	24 (illustrative)	Capability-profile parameter.
RES-015	Configuration backup interval	h	24 or on change (illustrative)	Capability-profile parameter.
RES-015	Restore test interval per class	months	12 (illustrative)	Capability-profile parameter.
RES-016	Checkpoint interval	min	Optimised from failure rate and checkpoint cost (illustrative 60)	Young/Daly-style approximation; capability-specific.
RES-016	Durable checkpoint copies outside cluster failure domain	count	1 (illustrative)	Capability-profile parameter.

Control	Parameter	Unit	Illustrative default	Basis
RES-016	Restart-from-checkpoint time	min	Within RTO (illustrative 30)	Derived from RES-001.
RES-017	Programme adherence target	% of scheduled tests completed on time	95 (illustrative)	Capability-profile parameter.
RES-018	Integrated test cycle	months	24 (illustrative)	Capability-profile parameter.
RES-018	Representative load during test	% of design IT load	As high as safely authorised (illustrative 70)	Load banks or real workload; site-specific.
RES-019	Tabletop cycle	months	6 (illustrative)	Capability-profile parameter.
RES-020	Maximum age of open gate-critical lesson	days	90 (illustrative)	Capability-profile parameter.
RES-021	Crisis command activation time	min from trigger	30 (illustrative)	Capability-profile parameter.
RES-023	Emergency change retrospective review deadline	working days	5 (illustrative)	Capability-profile parameter.
RES-024	Post-event review deadline	days	30 (illustrative)	Capability-profile parameter.

Note on checkpoint interval. For a job with checkpoint cost *C* (time to write a checkpoint) and mean time between job-interrupting failures *M*, a widely used first-order approximation for the interval that minimises expected lost time is √(2·C·M) (Young's approximation, refined by Daly). The capability owner should record *C* and *M* as measured for the Capability Instance and justify the chosen interval against the lost-work tolerance; the approximation is a guide, not a requirement.

10 Evidence expectations by claim type

Control	Claim type	Profile tier	Required classes	Quality floor	Sufficiency floor	Mandatory items
RES-001	DESIGN	CP-1	E1+E7	P3 F3 C2 R3 T2 I2	S3	Approved tolerance statement; Business-impact analysis
RES-003	DESIGN	CP-1	E1+E7	P3 F3 C2 R3 T2 I2	S3	Scenario library; Applicability rationale
RES-004	DESIGN	CP-1	E1+E7	P3 F3 C2 R3 T2 I2	S3	Appointment record; Responsibility matrix
RES-009	DESIGN	CP-1	E1+E7	P3 F3 C2 R3 T2 I2	S3	Approved strategy
RES-012	DESIGN	CP-1	E1+E7	P3 F3 C2 R3 T2 I2	S3	Approved priority scheme
RES-005	IMPLEMENTATION	CP-2	E1+E2+E3	P3 F3 C3 R3 T3 I2	S4	Fault-domain model; Reconciliation record; Shared-domain register
RES-006	IMPLEMENTATION	CP-2	E1+E2+E3	P3 F3 C3 R3 T3 I2	S4	Boundary register; Configuration or inspection evidence
RES-007	IMPLEMENTATION	CP-2	E1+E2+E7	P3 F3 C3 R3 T3 I2	S4	Blast-radius analysis; Exceedance register

Control	Claim type	Profile tier	Required classes	Quality floor	Sufficiency floor	Mandatory items
RES-011	IMPLEMENTATION	CP-2	E1+E2+E5	P3 F3 C3 R3 T3 I2	S3	Provider continuity evidence; Reconciliation
RES-013	IMPLEMENTATION	CP-2	E1+E2+E3	P3 F3 C3 R3 T3 I2	S5	Recovery dependency map; Circular-dependency analysis
RES-017	OPERATION	CP-3	E3+E4	P3 F3 C3 R3 T3 I3	S3	Test records; Defect records
RES-019	OPERATION	CP-3	E3+E6	P3 F3 C3 R3 T3 I3	S3	Exercise reports; Action tracker
RES-020	OPERATION	CP-3	E1+E3	P3 F3 C3 R3 T3 I3	S3	Lessons register; Retest evidence for closed gate-critical items
RES-022	OPERATION	CP-3	E1+E3	P3 F3 C3 R3 T3 I3	S3	Notification logs; Obligation register
RES-023	OPERATION	CP-3	E1+E2+E3	P3 F3 C3 R3 T3 I3	S3	Emergency change records; Retrospective reviews
RES-024	OPERATION	CP-3	E1+E3+E7	P3 F3 C3 R3 T3 I3	S3	Post-event reviews; Reassessment trigger records
RES-002	EFFECTIVENESS	CP-4	E2+E3+E4	P3 F4 C3 R4 T3 I3	S4	Degradation mode catalogue; Test or event records
RES-010	EFFECTIVENESS	CP-4	E2+E3+E4	P3 F4 C3 R4 T3 I3	S4	Relocation test record; Contract or reservation evidence
RES-015	EFFECTIVENESS	CP-4	E2+E3+E4	P3 F4 C3 R4 T3 I3	S4	Restore test records; Integrity verification evidence
RES-021	EFFECTIVENESS	CP-4	E1+E3+E4+E6	P3 F4 C3 R4 T3 I3	S4	Exercise/event records with timings; Decision logs
RES-008	RESILIENCE	CP-5	E1+E2+E3+E4	P4 F4 C4 R4 T3 I3	S5	Validation plan; Results for gate-critical boundaries
RES-014	RESILIENCE	CP-5	E1+E2+E3+E4	P4 F4 C4 R4 T3 I3	S5	Restoration sequence; Integrated test record with stage timings
RES-016	RESILIENCE	CP-5	E1+E2+E3+E4	P4 F4 C4 R4 T3 I3	S5	Checkpoint policy; Restart test record; Integrity verification logs
RES-018	RESILIENCE	CP-5	E1+E2+E3+E4	P4 F4 C4 R4 T3 I3	S5	Integrated test records; Safety case

Typical decisive evidence:

- **DESIGN** (RES-001, RES-003, RES-004, RES-009, RES-012): approved statements and analyses with accountable approval. Tolerances must be measurable and internally consistent (for example RPO compatible with checkpoint interval).
- **IMPLEMENTATION** (RES-005, RES-006, RES-007, RES-011, RES-013): models and registers reconciled to primary as-built sources (single-line diagrams, P&IDs, rack power and cooling maps, fabric topology, scheduler and storage placement). Reconciliation by a party other than the author.
- **OPERATION** (RES-017, RES-019, RES-020, RES-022, RES-023, RES-024): records across a representative period showing the programme runs as designed, with exceptions handled.
- **EFFECTIVENESS** (RES-002, RES-010, RES-015, RES-021): tests or events demonstrating the safeguard works under defined adverse conditions, with timings.
- **RESILIENCE** (RES-008, RES-014, RES-016, RES-018): scenario and recovery tests with cross-domain state coverage, independent witnessing and capability-level measurements.

Evidence pitfalls:

- **Plans without tests** support DESIGN only (AIDCF-008 Part XIII, CG-10 decision effect).
- **Component tests are not integrated tests.** A generator that starts on a load bank and a UPS that transfers on a monthly test do not demonstrate that an AI cluster survives utility loss; the interaction under AI load steps must be tested or analysed (RES-018).
- **System availability is not outcome restoration.** "All nodes up" does not demonstrate that training resumed at required throughput from a verified checkpoint.
- **Tabletop exercises evidence decision processes**, not technical recovery.

11 Assessment guidance

11.1 Start from tolerances

Obtain RES-001 tolerances first. Every later RES judgement compares something against them: degradation residual capacity, blast radius, alternate-capacity activation time, restoration duration, lost work.

11.2 Build or obtain the fault-domain and recovery-dependency graph

Where the organisation maintains an AIDCF-014 conformant model, query it for (a) redundancy groups whose members share a FaultDomain, and (b) cycles in recovery DEPENDS_ON relationships. Where it does not, construct a minimum version from as-built documents for the gate-critical outcomes before evaluating RES-005, RES-007 and RES-013.

11.3 Walk the restoration sequence

For RES-014, walk the sequence stage by stage with the people who would execute it. At each stage ask: what are the entry criteria, who decides, what credentials and tools are needed and where are they, how long did this take in the last test, and what is the hold point before the next stage? Cross-check against POW-024, THM-024, NET-015, STO-019 and CMP-023 runbooks; inconsistencies between domain runbooks are frequent and material.

11.4 Judge test representativeness

Integrated tests are often run at reduced load, on a subset of the estate, or with simulated components. That is acceptable if the representativeness argument is explicit: which behaviours scale linearly, which do not (load steps, thermal transients, subnet-manager sweeps, storage metadata recovery), and what residual UNKNOWN remains.

DG-RES-REQ-002 An assessor shall record, for every test used as RESILIENCE evidence, the load level, scale, operating states and components exercised, and shall record a U5 coverage-gap UNKNOWN for any gate-critical behaviour that the test did not exercise and that is not covered by analysis.

DG-RES-REQ-003 An assessor shall not accept a restoration sequence as tested end-to-end unless either a full test occurred or every hand-off between adjacent segment tests was itself tested.

11.5 Crisis management evidence

For RES-021 and RES-022, evidence should show that command activated within target time, decisions were taken by the right authority (including authority for controlled shutdown and tenant suspension), decision logs exist, and communications reached stakeholders within contractual and regulatory deadlines. Interview alternates, not only primary role holders.

11.6 Key assessment questions by family

- **RES-01 Resilience strategy.** Is there an approved, measurable impact tolerance per Capability Instance, derived from business-impact analysis rather than copied from a service catalogue? Are graceful-degradation actions fast enough relative to UPS autonomy and thermal ride-through, and have they been measured? Does the scenario library include maintenance-induced, recovery-induced and compound scenarios? Does one person hold accountable authority across facility, IT and provider boundaries?
- **RES-02 Fault-domain and containment design.** Has the fault-domain model been reconciled against primary as-built sources, and which redundancy claims fail the shared-domain test? Is blast radius expressed in outcome units and compared with tolerance, including firmware, configuration, identity and automation common modes? Which containment boundaries have been validated at V5 or above, and which only by inspection?
- **RES-03 Continuity planning.** Is the continuity strategy explicit about site loss, provider loss and control-plane loss? Is alternate capacity actually reserved or contracted on terms that hold during a regional surge, and has a relocation been tested end to end including checkpoint transfer? Are provider recovery objectives compatible with the tolerances? Are crisis priorities implemented in scheduler classes, not just written down?
- **RES-04 Recovery planning.** Which recovery prerequisites are hosted on the infrastructure they help recover, and how is each cycle broken? Does the restoration sequence bring cooling up before heat load and respect load-step limits? Has each hand-off between restoration stages been tested? Can configuration be restored from a pre-incident, integrity-verified baseline? Has a training job restarted from checkpoint on different nodes within RTO?
- **RES-05 Testing and exercising.** Does the component test programme run as scheduled, and are failures closed? Which prioritised scenarios have been integrated-tested at representative load, and what did the tests measure at capability level? Do tabletop exercises include cyber-physical and compound scenarios? Are gate-critical lessons closed only after retest?
- **RES-06 Crisis and adaptive response.** Can crisis command activate and communicate without site systems or corporate identity? Who is authorised to order controlled shutdown or tenant suspension, and do alternates know it? Are emergency changes reconciled back to baseline? Do post-event reviews raise reassessment triggers against affected controls and gates?

12 Testing safety

Resilience testing is the most hazardous testing in AIDCF. Integrated scenario tests, black-start tests, isolation validation of electrical protection, fire or fluid isolation, and degradation tests that remove power or cooling capacity involve hazardous energy, risk to equipment and risk to customer service.

DG-RES-REQ-004 No integrated scenario, black-start, isolation-validation or capacity-removal test shall be performed without written authorisation from the facility owner, the relevant electrical and mechanical engineering authorities and the capability owner; a safety case; permit-to-work for energised or hazardous work (OPS-007, OPS-008); a named abort authority with defined abort criteria; and a restoration plan.

DG-RES-REQ-005 Live capability exercises at AIDCF-004 validation level V7 shall additionally require executive authorisation and advance notice to affected tenants and customers.

DG-RES-REQ-006 Assessors shall not initiate, direct or modify any test; they may observe authorised tests and review records.

Preferred alternatives in order: analysis of real events with good telemetry; segmented tests with tested hand-offs; simulation and digital-twin analysis validated against component tests; tabletop walk-throughs to expose procedural gaps.

13 UNKNOWN hot-spots

Hot-spot	Typical type	Typical severity	Resolution route
Impact tolerance not defined or not approved	U1	US5	Capability owner approval (RES-001, CAP-010)
Fault-domain model not reconciled to as-built	U3/U5	US4 (CG-09)	Reconcile to SLD, P&ID, rack maps and topology
Circular recovery dependency not analysed	U10	US4 (CG-10)	Dependency cycle analysis; break documentation
Restoration sequence never tested end-to-end	U5	US4 (CG-10)	Segmented tests with tested hand-offs
Checkpoint restart never tested on alternate nodes	U5	US4 (CG-10)	Restart test at representative scale
Integrated test at unrepresentative load	U5	US3/US4	Representativeness analysis; higher-load test
Provider continuity arrangements undisclosed	U2	US3 (US4 if sole provider)	Contractual evidence rights; E5 assurance
Alternate capacity untested	U1	US3 (US4 if sole strategy)	Relocation test
Post-event review incomplete	U3	US3	Complete review; mark evidence STALE meanwhile

14 Interfaces with other domains

Domain	Interface	Key controls
CAP	Service Outcomes, criticality and impact tolerance	CAP-008, CAP-010
DEP	Recovery dependency mapping, shared sources and routes, substitution	DEP-004, DEP-005, DEP-007, DEP-016
POW	Electrical black-start, load-step response, UPS runtime	POW-011, POW-020, POW-024
THM	Resilient capacity, thermal recovery, isolation and drainage	THM-003, THM-020, THM-024
NET	Fabric fault domains, OOB resilience, configuration management, core services	NET-004, NET-015, NET-022, NET-024
CMP	Compute fault domains, rollback, node rebuild	CMP-008, CMP-016, CMP-023
STO	Checkpoint durability, restart readiness, storage recovery sequencing	STO-017, STO-018, STO-019
SEC	Break-glass, incident containment, safe rollback	SEC-004, SEC-019, SEC-024
OPS	Permit-to-work, incident response, change, temporary configuration	OPS-007, OPS-014, OPS-018, OPS-019
GOV	Accountability, stakeholder obligations, external reporting	GOV-005, GOV-006, GOV-019, GOV-022
EVA	Change triggers and evidence expiry after events	EVA-021, EVA-022

15 Informative references

Cited at document level; no clause-level text is reproduced.

- ISO 22301 (Business continuity management systems: requirements); ISO 22313 (guidance); ISO/TS 22317 (business impact analysis); ISO 22316 (organizational resilience); ISO 22320 (incident management).
- ISO/IEC 27031 (ICT readiness for business continuity).
- NIST SP 800-34 Rev. 1 (contingency planning); NIST SP 800-84 (test, training and exercise programmes); NIST SP 800-61 Rev. 3 (incident response).
- IEC 31010 (risk assessment techniques); IEC 60812 (FMEA); IEC 61025 (fault tree analysis).
- Regulation (EU) 2022/2554 (Digital Operational Resilience Act); UK PRA/FCA operational resilience policy.
- ANSI/TIA-942; EN 50600 series / ISO/IEC 22237 series.
- ISA/IEC 62443-2-1 (for OT configuration backup and restoration context).
- AIDCF-004, AIDCF-008, AIDCF-010, AIDCF-013, AIDCF-014.

16 Control summary

Control ID	Title	Control type	Required AL	Claim type	Gate mappings
MCF-RES-001	Minimum Service Outcome	DIRECTIVE	AL2	DESIGN	CG-01 SUPPORTING; CG-10 SUPPORTING; CG-09 SUPPORTING
MCF-RES-002	Graceful Degradation	CORRECTIVE	AL5	EFFECTIVENESS	CG-09 SUPPORTING; CG-10 SUPPORTING
MCF-RES-003	Resilience Scenarios	DIRECTIVE	AL2	DESIGN	CG-09 SUPPORTING; CG-10 SUPPORTING; CG-03 SUPPORTING
MCF-RES-004	Resilience Ownership	DIRECTIVE	AL2	DESIGN	CG-13 SUPPORTING
MCF-RES-005	Fault-Domain Definition	PREVENTIVE	AL3	IMPLEMENTATION	CG-09 CRITICAL; CG-03 SUPPORTING; CG-02 SUPPORTING
MCF-RES-006	Containment Boundaries	PREVENTIVE	AL3	IMPLEMENTATION	CG-09 SUPPORTING
MCF-RES-007	Blast Radius	PREVENTIVE	AL3	IMPLEMENTATION	CG-09 CRITICAL; CG-03 SUPPORTING
MCF-RES-008	Isolation Validation	DETECTIVE	AL5	RESILIENCE	CG-09 CRITICAL; CG-04 SUPPORTING
MCF-RES-009	Continuity Strategy	DIRECTIVE	AL2	DESIGN	CG-10 SUPPORTING; CG-01 SUPPORTING
MCF-RES-010	Alternative Capacity	RECOVERY	AL5	EFFECTIVENESS	CG-10 SUPPORTING; CG-03 SUPPORTING
MCF-RES-011	Site and Provider Continuity	PREVENTIVE	AL3	IMPLEMENTATION	CG-03 SUPPORTING; CG-10 SUPPORTING
MCF-RES-012	Crisis Priorities	DIRECTIVE	AL2	DESIGN	CG-10 SUPPORTING; CG-01 SUPPORTING
MCF-RES-013	Recovery Dependencies	RECOVERY	AL3	IMPLEMENTATION	CG-10 CRITICAL; CG-03 SUPPORTING
MCF-RES-014	Recovery Sequencing	RECOVERY	AL5	RESILIENCE	CG-10 CRITICAL; CG-05 SUPPORTING; CG-06 SUPPORTING
MCF-RES-015	Configuration Restoration	RECOVERY	AL5	EFFECTIVENESS	CG-10 SUPPORTING; CG-07 SUPPORTING; CG-14 SUPPORTING
MCF-RES-016	Data and Checkpoint Recovery	RECOVERY	AL5	RESILIENCE	CG-10 CRITICAL; CG-08 SUPPORTING

Control ID	Title	Control type	Required AL	Claim type	Gate mappings
MCF-RES-017	Component Tests	DETECTIVE	AL4	OPERATION	CG-10 SUPPORTING; CG-05 SUPPORTING; CG-06 SUPPORTING
MCF-RES-018	Integrated Scenario Tests	DETECTIVE	AL5	RESILIENCE	CG-10 CRITICAL; CG-09 SUPPORTING; CG-05 SUPPORTING; CG-06 SUPPORTING
MCF-RES-019	Tabletop Exercises	DETECTIVE	AL4	OPERATION	CG-10 SUPPORTING; CG-13 SUPPORTING
MCF-RES-020	Lessons and Retest	CORRECTIVE	AL4	OPERATION	CG-10 SUPPORTING; CG-14 SUPPORTING
MCF-RES-021	Crisis Command	CORRECTIVE	AL5	EFFECTIVENESS	CG-13 SUPPORTING; CG-10 SUPPORTING
MCF-RES-022	Stakeholder Communications	CORRECTIVE	AL4	OPERATION	CG-13 SUPPORTING
MCF-RES-023	Emergency Change	CORRECTIVE	AL4	OPERATION	CG-14 SUPPORTING
MCF-RES-024	Post-Event Adaptation	CORRECTIVE	AL4	OPERATION	CG-14 SUPPORTING

17 Worked synthetic example

This example is entirely synthetic. The site, figures and results are invented to illustrate the method and do not describe any real facility, organisation or test.

Context. Synthetic site S-2 hosts Capability Instance CI-T2 (a 4,096-accelerator training cluster, liquid-cooled) and CI-I2 (an inference service on a separate air-cooled hall). Decision question: can CI-T2 be restored to ≥ 90% of nominal training throughput within 12 hours of a site-wide power loss with no more than 2 hours of lost training progress, as of the assessment date?

Tolerances (RES-001). Approved: maximum tolerable disruption 24 h; RTO 12 h; maximum lost training progress 2 h; minimum retained capacity in DEGRADED state 50%. Checkpoint interval 45 min, justified with measured checkpoint cost of 90 s and observed job-interrupting failure interval (synthetic) using the Young/Daly approximation.

Assessment findings.

Control	Claim type	Evidence	Result
MCF-RES-005	IMPLEMENTATION	Fault-domain graph reconciled to SLD, P&ID and rack maps; one CDU pair found fed from the same switchboard section	CONFORMING (shared domain recorded and dispositioned: a second feed was already in the approved change plan and is captured as an accepted limitation with expiry)
MCF-RES-007	IMPLEMENTATION	Blast-radius analysis covers 18 prioritised scenarios; switchboard-section loss removes 50% of CI-T2 cooling, within the 50% DEGRADED minimum	CONFORMING
MCF-RES-008	RESILIENCE	CDU isolation and electrical discrimination validated at V5; fabric partition validated at V5; fire-compartment cable separation validated only at V1 inspection	PARTIAL (Finding: V1 insufficient for a gate-critical boundary without justification)

Control	Claim type	Evidence	Result
MCF-RES-013	IMPLEMENTATION	Recovery dependency graph; cycle detection found the image registry for node rebuild hosted on the CI-T2 storage cluster, with no offline copy	NONCONFORMING (cycle without break)
MCF-RES-014	RESILIENCE	Segmented tests: power-to-cooling (site test) and network-to-control-plane (cluster test); hand-off from cooling-stable to staged node power-on not tested	UNKNOWN (U5, US4)
MCF-RES-016	RESILIENCE	Restart test on alternate nodes completed in 22 min; checkpoint integrity verified	CONFORMING
MCF-RES-018	RESILIENCE	Utility-loss test at 60% IT load witnessed independently; jobs checkpointed and drained as designed; representativeness analysis covers load steps	CONFORMING

Gate roll-up (SPEC §6.2).

- **CG-09**: CRITICAL controls RES-005 and RES-007 are CONFORMING; RES-008 is PARTIAL → **FAIL** (also subject to NET-004 and other domains' CRITICAL controls).
- **CG-10**: RES-013 is NONCONFORMING → **FAIL**, irrespective of RES-014 being UNKNOWN and RES-016 and RES-018 being CONFORMING.

Decision effect and actions. The decision recommendation is bounded by the AIDCF-010 ceilings for CG-09 and CG-10 FAIL; the US4 UNKNOWN on RES-014 is also carried to CG-12. Actions: place an offline, integrity-verified copy of node images and firmware bundles outside the CI-T2 storage fault domain and re-run cycle detection; test the cooling-stable to staged power-on hand-off with load-step measurements under authorisation; justify or raise the validation level for fire-compartment separation (for example through a fire-engineering review and inspection of cable route separation). Each action carries a reassessment trigger linked to the affected controls and gates.

What the example illustrates. The site had strong component and integrated test evidence and a conforming checkpoint regime, yet recovery readiness failed on a single unbroken circular dependency. This is typical: RES assessments most often fail not on whether systems can be restarted, but on whether the things needed to restart them survive the event.